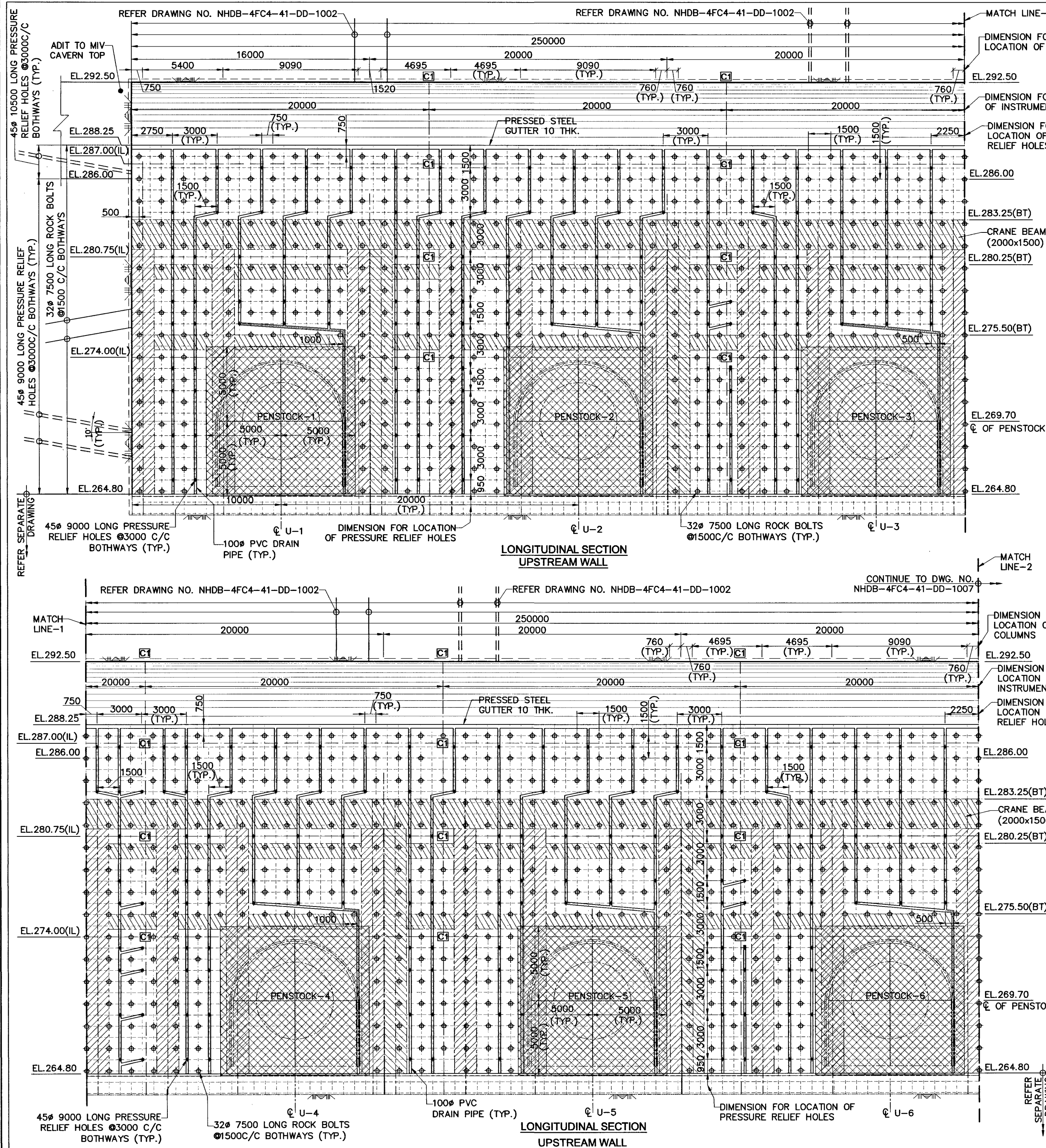




NOTES FOR INSTRUMENTATION :-

- MINOR MODIFICATION IN THE NUMBERS AND LOCATIONS OF INSTRUMENTS MAY BE DONE AS THE EXCAVATION PROGRESSES TO SUIT SITE CONDITIONS AS PER THE DIRECTION OF ENGINEER-IN-CHARGE.
- IMMEDIATELY AFTER INSTALLATION OF ANY INSTRUMENTS, ITS POSITION AND LEVEL SHALL BE PRECISELY SURVEYED. THE LEVEL AND CO-ORDINATE SHALL BE COMPUTED AND SUBMITTED TO THE ENGINEER-IN-CHARGE.
- INSTRUMENTS SHALL BE INSTALLED AS PER MANUFACTURER'S INSTRUCTION MANUAL AND AS SPECIFIED IN SECTION B.18 OF TECHNICAL SPECIFICATION.
- ELECTRICAL VIBRATING WIRE TYPE INSTRUMENTS SHALL BE SUITABLY CONNECTED TO DATA ACQUISITION SYSTEM ENCASED IN CONTROL BLOCK OF POWER HOUSE. FURTHER ARRANGEMENT SHALL INCLUDE, WHERE NECESSARY, THE PROVISION OF TEMPORARY READ OUT POINTS.
- ANY DAMAGED INSTRUMENT/CABLE DURING THE PROGRESS OF THE WORK SHALL BE REPAIRED OR REPLACED IMMEDIATELY.
- THE CONTRACTOR SHALL SUBMIT MONTHLY REPORT OF INSTRUMENTATION BASED ON DAILY READINGS OF ALL INSTRUMENTS IN SPECIFIED FORMAT ALONG WITH ANALYSIS OF OBSERVED DATA UNLESS STATED OTHERWISE OR DIRECTED BY ENGINEER-IN-CHARGE IN REGARD TO FREQUENCY OF READINGS.
- THE MONITORING OF INSTRUMENTATION SHALL BE STARTED IMMEDIATELY AFTER INSTALLATION CONFORMING TO SECTION B.18 OF TECHNICAL SPECIFICATION AND DATA SHALL BE PROVIDED TO DESIGN DIVISION MONTHLY.
- ALL THE INSTRUMENTS ARE TO BE PLACED ALONG THE SECTION LINES AS SHOWN IN THE DRAWING.

NOTES FOR POST TENSIONED ROCK BOLTS (TENDONS):-

- TENDONS, THEIR INSTALLATION, TENSIONING AND TESTING SHALL CONFORM TO SECTION B.4 OF TECHNICAL SPECIFICATIONS.
- THE ENTIRE WORK OF DRILLING, GROUTING, FABRICATION, INSTALLATION AND TESTING OF TENDONS SHALL BE CARRIED OUT BY A SPECIALIST AGENCY HAVING EXPERIENCE, EQUIPMENT AND TECHNOLOGY FOR EXECUTION OF THESE WORKS AND UNDER STRICT QUALITY CONTROL.
- THE CONTRACTOR SHALL SUBMIT COMPLETE DETAILS CONCERNING TYPE AND MATERIALS OF THE TENDONS TOGETHER WITH MANUFACTURERS INSTRUCTIONS AND CERTIFICATIONS, METHODS OF INSTALLATION, CORROSION PROTECTION, STRESSING EQUIPMENT, GROUTING METHOD AND EQUIPMENT, PROPOSAL FOR SUITABILITY AND STRESSING TESTS AND EQUIPMENT.
- TYPES OF POST TENSIONED ROCK BOLTS (TENDONS):-
(a) TYPE-2: CONSISTING OF HIGH STRENGTH STEEL TENDON FITTED ON HEAD ON THE ROCK SURFACE AT BOTH END, WITH ONE END FOR ANCHORAGE, A MEANS PERMITTING FORCE TO BE TRANSFERRED, AND OTHER END FOR STRESSING.
- CONSOLIDATION GROUTING, WHEREVER REQUIRED SHALL BE CARRIED OUT PRIOR TO INSTALLATION OF TENDONS.

TABLE OF TENDONS:-

SL. NO.	TYPE	LENGTH (M)	NOS.	TOTAL LENGTH (M) (TENTATIVE)
1.	T2			TENDONS DETAILS (LENGTH, NOS., TOTAL LENGTH) REMAINS SAME AS SHOWN IN DWG. NO. NHDB-4CC4-41-DD-1012

REFERENCES:-

- NHDB-4AC4-41-DD-0001
- NHDB-4AC4-41-DD-1003
- NHDB-4FC4-41-DD-1002
- NHDB-4FC4-41-DD-1004
- NHDB-4FC4-41-DD-7001
- NHDB-4BC4-41-DD-1038 & 1041
- NHDB-4CC4-41-DD-1012 TO 1025

GENERAL NOTES:

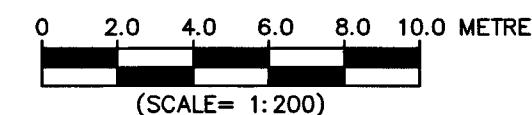
- सभी माप मिलीमीटर में तथा ऊंचाई मीटर में दिए गए हैं।
- ALL DIMENSIONS ARE IN MILLIMETRE AND ELEVATIONS ARE IN METRE.
- THIS DRAWING IS ISSUED FOR ROCK SUPPORTS, CABLE TENDONS, PRESSURE RELIEF HOLES & INSTRUMENTATION TYPICAL ARRANGEMENT IN WALLS OF MIV CAVERN FROM EL.288.25 TO EL.264.80.
- FOR DETAILS OF ROCK SUPPORTS BELOW EL.264.80 REFER SEPARATE DRAWINGS.
- THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWINGS NO. NHDB-4FC4-41-DD-1002 & NHDB-4AC4-41-DD-1003.
- THIS DRAWING SHALL BE READ IN CONJUNCTION WITH RELEVANT JUNCTION EXCAVATION DRAWING OF ADITS/PENSTOCKS WITH MIV CAVERN WALL.
- THE DEFINITION OF 'A' AND 'B' LINE SHALL BE AS PER SECTION B.3 OF TECHNICAL SPECIFICATIONS.
- THE LENGTH & SPACING OF ROCK BOLTS SHOWN ARE INDICATIVE AND MAY BE MODIFIED WITH THE APPROVAL OF ENGINEER-IN-CHARGE TO SUIT THE SITE CONDITIONS.
- THE DRAINAGE PIPE NETWORK SHOWN IS SCHEMATIC AND MAY BE MODIFIED TO SUIT THE SITE REQUIREMENTS.
- THE PVC PIPE SHALL CONFORM TO SECTION B-16 OF TECHNICAL SPECIFICATION.
- THE PRESSURE RELIEF HOLES SHALL BE CLEANED BEFORE INSTALLATION OF DRAINAGE PIPE.
- CONSOLIDATION GROUTING, WHEREVER REQUIRED SHALL BE CARRIED OUT PRIOR TO INSTALLATION OF PRESSURE RELIEF HOLES.
- THE LOCATION OF ROCK BOLTS AND PRESSURE RELIEF HOLES SHALL BE ADJUSTED TO SUIT THE LOCATIONS OF TENDONS AND INSTRUMENTS WITH THE APPROVAL OF ENGINEER-IN-CHARGE.
- THIS DRG. SHALL BE REFER FOR INSTRUMENTS BETWEEN EL.288.25 TO EL.264.80. FOR INSTRUMENTATION OF MIV CAVERN CROWN REFER DRG. NO.-NHDB-4FC4-41-DD-7001.

TABLE OF INSTRUMENTS:-

SL. NO.	NAME OF INSTRUMENT	SYMBOL	NOS.
1.	CONVERGENCE MEASUREMENT (OPTICAL TARGETS)	×	77
2.	LOAD CELL (30T CAPACITY)	⊙	77
3.	MULTIPLE POINT BORE HOLE EXTENSOMETER (5M, 10M, 15M) (3 POINTS)	⦿	77

LEGENDS:-

- ⊕ 32# 7500 LONG ROCK BOLTS
- ⊕ 45# 9000/10500 LONG PRESSURE RELIEF HOLE
- ×
- ⊙
- ⦿
- ⊕ ⊗ ⊙ (COMBINATION OF INSTRUMENTS-1)
- BT BEAM TOP LEVEL
- IL INSTRUMENTATION LEVEL
- ⊗ ROCK SUPPORTS AND PRESSURE RELIEF HOLES IN MARKED ZONE SHALL BE INSTALLED AS PER RELEVANT JUNCTION DRAWINGS OF INTERFACING STRUCTURE.



एन एच पी सी लिमिटेड
(एन एच पी सी लिमिटेड)
NHPC Limited
(A GOVT. OF INDIA NAVRATNA ENTERPRISE)
दिवांग बहुउद्देशीय परियोजना
DIBANG MULTIPURPOSE PROJECT

MIV CAVERN
ROCK SUPPORTS, CABLE TENDONS,
PRESSURE RELIEF HOLES AND INSTRUMENTATION
TYPICAL ARRANGEMENT
WALL ELEVATION

SHEET 1 OF 5

20.02.2026	00	ISSUED FOR CONSTRUCTION	BY: [Signature]	CH: [Signature]	APP: [Signature]	DATE: FEB. 2026	DRG. NO. NHDB 4FC4 41 DD 1006	00
DATE	NO.	REVISION OR ISSUES	BY	CH	APP	DATE	DRG. NO.	